

ROAN HOWARD

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EDUCATION

Stanford University, Stanford, CA

June 2027

BS in Computer Science, specializing in Artificial Intelligence; minor in Art Practice

Relevant Coursework: Linear Algebra and Multivariable Calculus (MATH51); Mathematical Foundations of Computing (CS103); Mechanics (PHYSICS41); Statistical Methods (STATS60); Probability for Computer Scientists (CS109); Programming Methodology (CS106A); Programming Abstractions (CS106B); Computer Organization and Systems (CS107); Information Science and Engineering (ENGR76); Design and Analysis of Algorithms (CS161); Natural Language Processing & Large Language Models (CS124); Matrix Methods (ENGR108); Continuous Mathematical Methods with an Emphasis on Machine Learning (CS205L); Machine Learning (CS229)

Student Organizations: Stanford Alaska Native Student Association; The American Indian Science and Engineering Society, Stanford Chess Club; Stanford Orientation Volunteer (Logistics for 1,500+ students); TreeHacks (acceptance rate ~7%); Kappa Sigma Fraternity, Intramural Basketball, Stanford American Indian Association, Stanford Powwow

Inglemoor High School, Kenmore, WA

June 2023

The International Baccalaureate Diploma Program; Chess Club President; Economics Club; Inglemoor High School Honors Society; Varsity Athlete: Soccer, Cross Country, Crew; Orientation Leader; Eastside Native American Program Coordinator

RECENT PROJECTS (more on website)

NewZ

April 2026

- NewZ turns the footage people are already recording into multi-angle local news, automatically, no editing, no captions, no profiles. Just point, shoot, and AI does the rest
- HackTech 2026 (Caltech), Consumer Product Winner, built using Python, TypeScript, and a multi-agent pipeline Gemini, Marengo, Claude

Hubbard Brook Agent

April 2026

- Built and deployed a full-stack AI agent featuring an animated Abenaki figure that reacts in real time to live Hubbard Brook sensor data, submitted to the 2026 Hubbard Brook Data Jam Art category
- Engineered a multi-tool Claude Sonnet backend using FastAPI, Anthropic SDK, Next.js, enabling natural language queries over 60 years of environmental records with live weather retrieval and historical trend analysis

Movie Service Agent

February 2026

- Engineered a multi-tool LLM agent leveraging DSPy ReAct, Mixtral-8x7B, and cosine-similarity collaborative filtering for real-time movie recommendations and ticket booking
- Augmented agent with RAG via SerpAPI retrieval and Mem0 semantic memory using Qdrant vector store and Alibaba-NLP dense embeddings on Together AI

Robot Data Labeler

February 2026

- In 12 hours at Stanford's annual hackathon, built a video-reasoning robotics referee in Python using Cosmos Reason, evaluating robot demonstrations for the Stanford IRL Lab dataset

RECENT WORK EXPERIENCE

SideShift, Full-Stack Engineer, New York, NY

June 2026 - Present

- Shipped 20+ features serving 1M+ creators and 4,000+ brands
- Delivered full-stack engineering work across search, sorting, user tooling, optimizations and end-to-end production pipelines, with continuous debugging and refactoring

Agentic AI Research, Researcher, Stanford, CA

March 2026 - June 2026

- Under PhD student Alexander Johansen, trained an MLP to replace a hardcoded sparsity parameter with query-adaptive prediction inside a Koopman operator-based multi-agent LLM retrieval system

New England Seafood International, Learning and Development Intern, London, UK

June 2025 - August 2025

- Revamped learning and development strategy for 600+ employees
- Deployed a Python script to automate Dayforce course uploads, saving 100+ hours of manual administrative time

SKILLS AND INTERESTS

- *Programming Languages:* Python, Julia, C++, C, Assembly, LaTeX, HTML, CSS, R, SQL, MATLAB
- *Libraries:* NumPy, Pandas, Matplotlib, scikit-learn, SciPy, TensorFlow, PyTorch, Keras and more
- *Tools:* Jupyter, Hugging Face, Nebius, Supabase, Vercel, GitHub, pgAdmin, DaVinci Resolve
- *Hobbies:* Ran 2025 Vancouver Marathon, author of 300+ film reviews, backpacked 15+ countries, learning German, sculptor